

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 3 – 6 February 2026 ¹

Problems:

1. **(2022 exam, Q1)** Consider a single-period model with N risky stocks and $\Omega = \{\omega_1, \omega_2, \dots, \omega_K\}$. Let $S_n(t)(\omega)$ denote the value of the n th stock at time point t in state ω . Assume there is a bank account with interest rate r .
 - (a) Explain the interpretation of a trading strategy $H = (H_0, H_1, \dots, H_N)$. In particular, what does it mean if H_i is negative for some $i \geq 0$?
 - (b) Assume that $N = 2$, $K = 3$ and $r = 0.2$ and that the price processes of the two stocks are given by:

n	$S_n(0)$	$S_n(1)(\omega_1)$	$S_n(1)(\omega_2)$	$S_n(1)(\omega_3)$
1	10	14.4	10.8	12
2	C	15.6	14.4	9.6

- i. For an arbitrary trading strategy H , determine the value process V and gains process G , and their discounted versions V^* and G^* .
- ii. For $\mathbb{Q} = (q_1, q_2, q_3)$, state a system of equations that the q_i must satisfy in order for $\mathbb{Q}$ to be a risk-neutral probability measure for this model. Determine for which values of C there exists a solution to this system of equations. For values of C too small for a solution to exist, explicitly describe an arbitrage strategy.
- iii. State a necessary condition for the existence of a risk-neutral probability measure, in terms of the intersection of G^* with a certain subset of $\mathbb{R}^3$.
- iv. If $C = 10$, use the risk-neutral valuation principle to find the fair prices of:
 - A. a European call option with strike price £13 based on the first stock,
 - B. a European put option with strike price £13 based on the second stock.
- (c) Let $0 < d < 1 < u$ and consider a model with just one stock, ie $N = 1$. Assume that the value of the stock at time point 0 is S_0 . At time point 1, the value of the stock is either $S_1 = dS_0$ or $S_1 = S_0$ or $S_1 = uS_0$, each with strictly positive probability p_1, p_2 and p_3 respectively.
 - i. Find values of r for which the model is free of arbitrage.
 - ii. Find values of r for which the model is complete.

You need to prove/justify your answers.

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